The Enterprise Continuum (EC) & Tools

Module 4

Module Objectives

- To provide an introduction to the Enterprise Continuum
- The purpose of the Enterprise Continuum
- The constituent pieces of the Enterprise Continuum
- To explain high-level issues with Tool Standardization

Overview

- A continuous extent of something, no part of which is different from any other
- A model for structuring a **virtual repository** and methods for classifying architecture and solution artifacts
- Based on architecture and solutions
  - Models, patterns, architecture descriptions
  - Deliverables in the ADM iterations
  - Assets from industry
  - Showing how artifacts evolve
- Physical implementation is the **Architecture Repository**

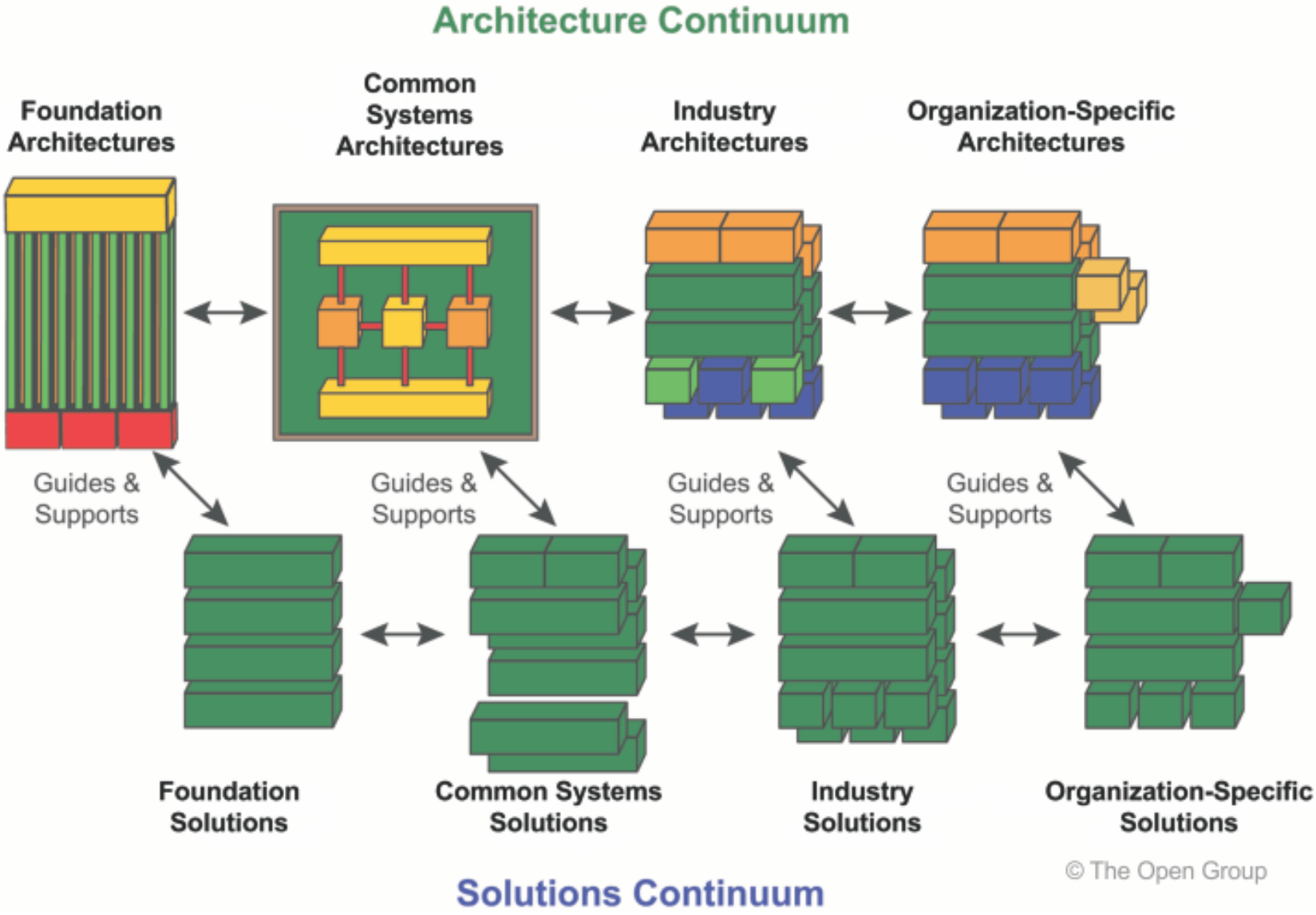
Overview (Cont'd)

- It is a combination of the **Architecture** and **Solutions** Continuum
- It enables effective use of COTS products
- It improves engineering efficiency
- It aids organization of reusable architecture and solution assets
- It provides a common language
  - within enterprises
  - between customer enterprises and vendors

Architecture Reuse

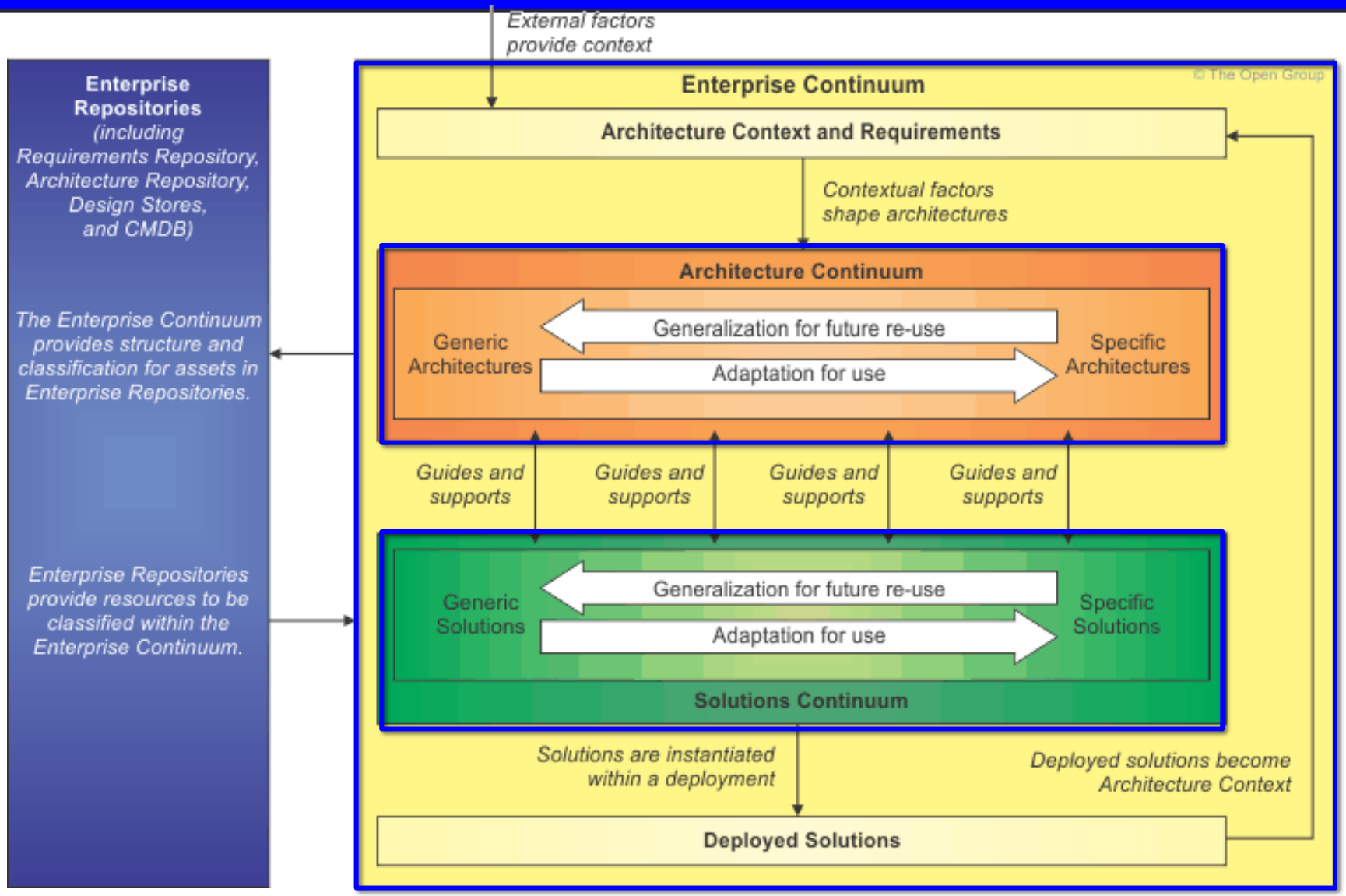
- The EC consists of **all architecture assets**: models, patterns, architecture descriptions, etc.
- **External assets** include
  - Generic reference models (TRM, Zachman…)
  - IT specific models (SOA, etc.)
  - Information Processing specific models (ecommerce etc.)
  - Vertical industry specific models (IBM IFW, etc.)
- The **architecture governance** function decides which assets should be part of EC

The Enterprise Continuum



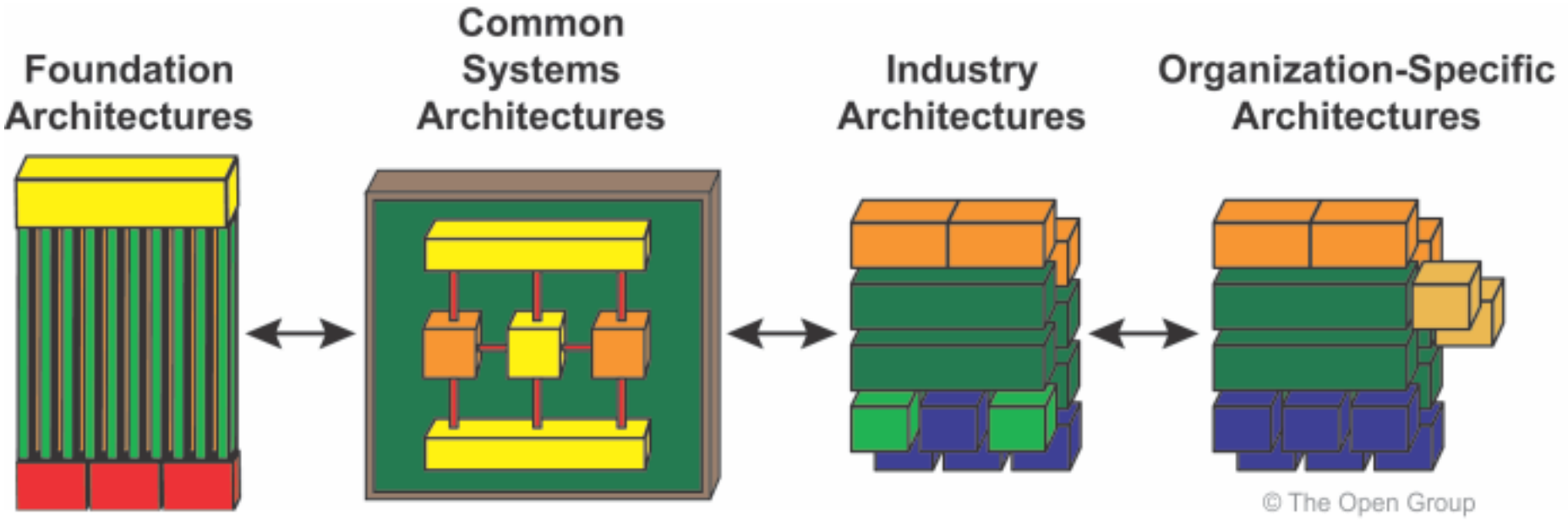
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Enterprise Continuum : Constituents

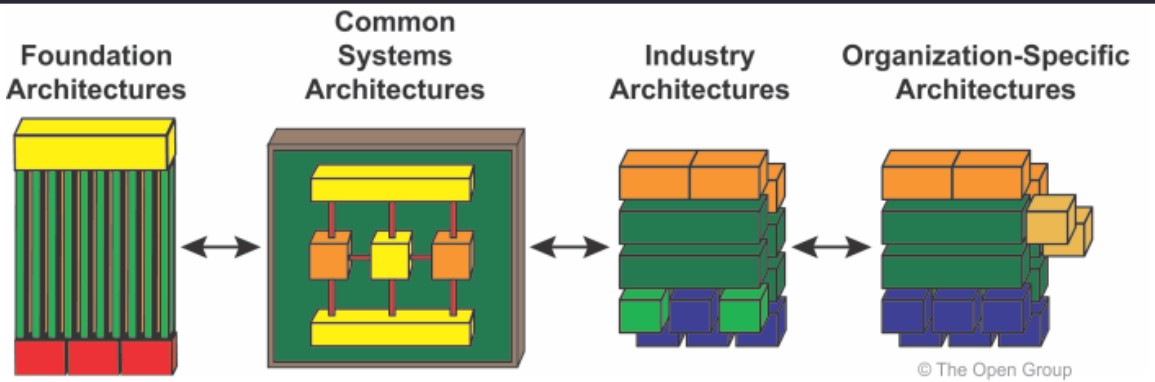


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The Architecture Continuum

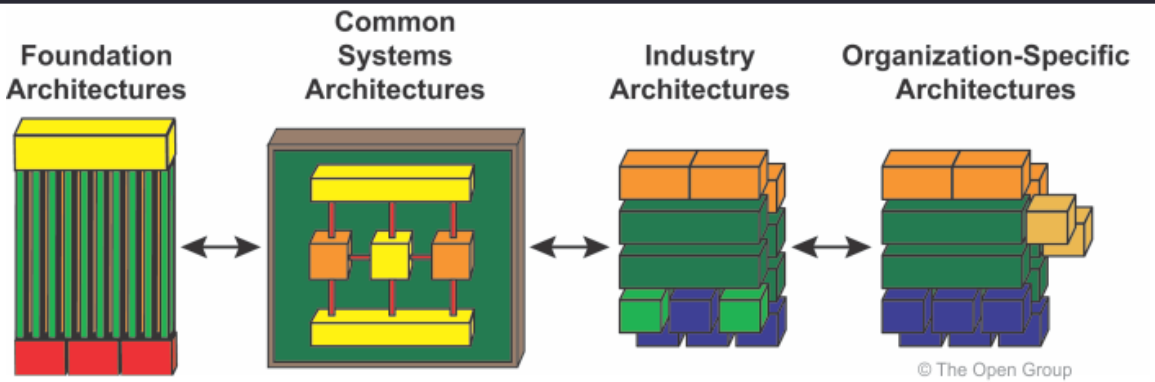


The Architecture Continuum



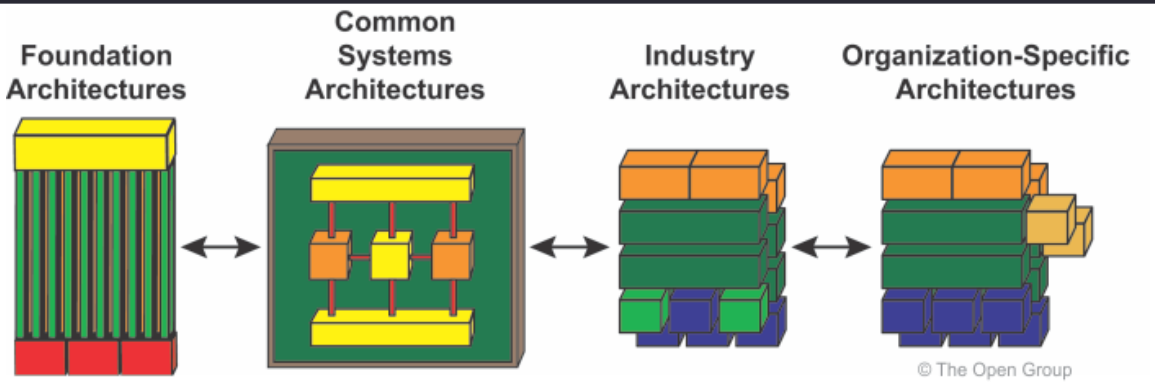
- Architecture range from **Foundation** through **Common Systems**, and **Industry** to an enterprise's own **Organization-Specific** architecture
- Arrows represent bi-directional relationship
  - **left to right**: meeting enterprise needs and business requirements
  - detail increases from left to right
  - **right to left**: leveraging architectural components and building blocks

The Architecture Continuum



- The architectural elements furthest left are the most reusable
- Requirements for missing elements are passed to the left of the continuum for inclusion
- Enterprises can use the same continuum models, specialized for specific businesses
- The figure shows different architectures, others may exist

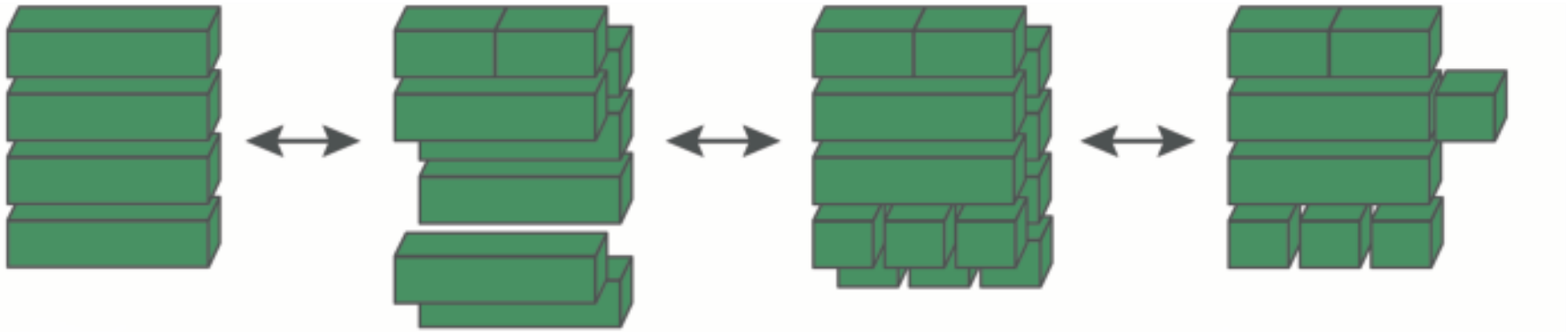
The Architecture Continuum



- Doesn't represent a formal process but a progression occurring at several levels
  - **Logical** to **Physical**
  - Horizontal (**IT focused**) to Vertical (**Business focused**)
  - **Generalization** to **Specialization**
  - **Taxonomy** to **Architecture Specification**
- At each point, an architecture is designed in terms of the design concepts and building blocks available

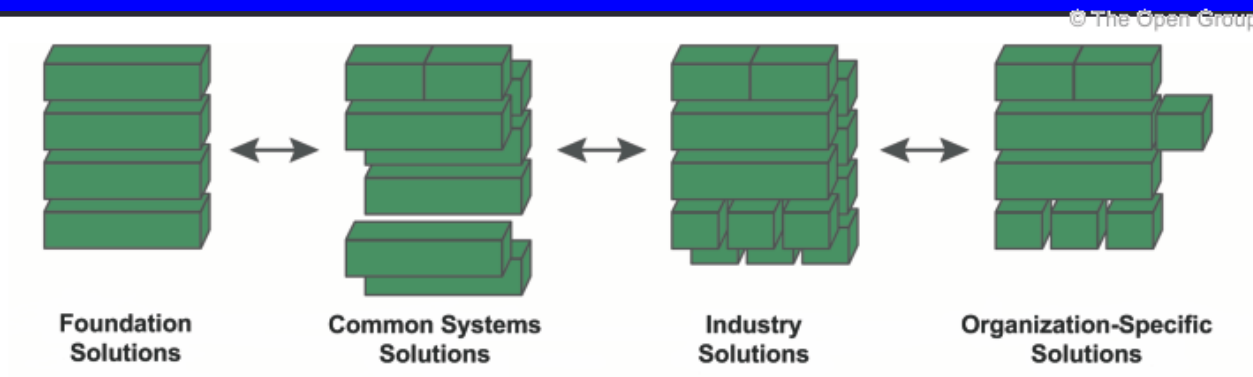
The Solutions Continuum

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The Solutions Continuum

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- The most specific architectures are on the right
  - Foundation solutions help to create common systems solutions
  - Common systems solutions help to create industry solutions
  - Industry solutions help to create organization specific solutions
- The most generic concepts are on the left
- The entire spectrum is important when balancing cost and value

The Solutions Continuum

- Represents the implementations of the architectures at the corresponding levels of the **Architecture Continuum**
- It is a population of the architecture with **Solution Building Blocks**, either purchased products or built components, that represent a solution to the enterprise's business need
- Forms a **Solutions Inventory** or **Reuse Library**, which adds significant value to the task of managing and implementing improvements to the IT environment

Relationships

- The **Architecture and Solutions Continuum** are related by guidance, direction and support
- E.g. the Foundation Architecture
  - is an architecture of building blocks and corresponding standards
  - supports all the Common Systems Architectures and therefore, the complete enterprise operating environment
- **TRM** is a Foundation Architecture
- **III-RM** is a Common Systems Architecture

Using the Continuum

- The TOGAF ADM describes the process of developing an enterprise-specific architecture by adopting and adapting generic architectures and solutions

- The Continuum
  - contains **complete** and **work-in progress** solutions
  - is a framework within a framework
  - has few internal assets, at first
  - grows by adding reusable building blocks

Relationships

- The **Solutions Continuum** assists understanding of products, systems, services and solutions
- The **EC** improves productivity through leverage
- The **EC** doesn't represent strictly changed relationships
  - enterprise architects may have components from a Common Systems Architecture
  - enterprise solutions may contain a product or a service

The Need for Tools

- Tools are needed to manage and control the artifacts within the EC
  - To promote re-use
  - To enable sharing of architecture information
  - To facilitate easier maintenance of the architecture
  - To ensure common terminology is used
  - To provide stakeholders with relevant models

Issues in Tools Standardization

- A single **"one size fits all"** tool versus multiple tools
- Can a single tool address all needs, all maturity levels?
- The Open Group recognizes the complexity in this area and is developing a TOGAF 9 Tools Certification program to assist with the evaluation.

Summary

- The Enterprise Continuum is
  - a model for structuring a **virtual repository** and methods for classifying architecture and solution artifacts
  - it enables the organization of **reusable architecture** and solution assets
  - it is also an aid to communication between all architects through **common language and terminology**
  - this in turn enables efficiency in engineering and effective **use of COTS products**

Summary

- **The Enterprise Continuum**
  - provides an overall context for architectures and solutions and classifies assets that apply across the entire scope of the enterprise
- **The Architecture Continuum**
  - provides a classification mechanism for assets that collectively define the architecture at different levels of evolution from generic to specific
- **The Solutions Continuum**
  - provides the classification for assets to describe specific solutions for the organization that can be implemented to achieve the intent of the architecture